TECTONICS, VOL. 5, NO. 1, PAGES 135-150, FEBRUARY 1986

PREFOLDING AND PREMEGAKINKING MAGNETIZATIONS FROM THE DEVONIAN COMERONG VOLCANICS, NEW SOUTH WALES, AUSTRALIA, AND THEIR BEARING ON THE GONDWANA POLE PATH

P. W. Schmidt and B. J. J. Embleton

CSIRO Division of Mineral Physics and Mineralogy, North Ryde, New South Wales, Australia

T. J. Cudahy and C. McA. Powell

Australian Plate Research Group, School of Earth Sciences, Macquarie University, North Ryde, New South Wales, Australia

Abstract. Palaeomagnetic results from the late Middle to early Late Devonian Comerong Volcanics in the Budawang Synclinorium, Lachlan Fold Belt (LFB), New South Wales, Australia, satisfy a fold test and support the hypothesis that foliated parts of the LFB have been folded by megakinks during the terminal mid-Carboniferous orogenic phase. The preferred pole position is at 76.9°S, 330.7°E (A_{95} = 7.2°), allowing calibration of the Australian pole path at around 370 Ma ago. An alterative pole position at 71.3°S 283.1°E (same A_{95}) takes into account a possible rotation of the megakinked terrane by 15° clockwise with respect to the Australian craton. Although there are still problems unravelling previously published results from Silurian rocks in the LFB, both because of unreliability and uncertainty as to where the LFB crustal units lay in relation to the Australian craton, there is no geological reason to suspect any large relative movement between the LFB and the craton since the Middle Devonian, and thus the Comerong Volcanics pole position is representative of the Late Devonian for Australia. The new pole requires reassessment of many other poles that have previously been accepted as representing the Siluro-Devonian for Australia, and Gondwanaland. We present reasons for believing that the magnetization, or remagnetization of many rock units (e.g., Mereenie Sandstone and Mulga Downs Group, Australia, and the Gneigura Supergroup, Africa) was later than sedimentation, probably during widespread Early Carboniferous deformation. Another rock unit (Msissi Norite, Africa) could be of Early Carboniferous, rather than Late Devonian, age.

Copyright 1986
by the American Geophysical Union.

Paper number 5T0705.
0278-7407/86/005T-0705$10.00

INTRODUCTION

The Palaeozoic palaeomagnetic results from Gondwanaland comprise predominantly those from Australia, particularly in the Silurian and Devonian [Embleton, 1981]. The Australian data can be interpreted in several ways depending upon age constraints and reliabilities assigned to certain pole positions. Four different Palaeozoic apparent polar wander paths (APWP's) have so far been proposed for Australia and, by implication, for Gondwanaland [Embleton et al., 1974; McElhinny and Embleton, 1974; Schmidt and Morris, 1977; Morel and Irving, 1978; Goleby, 1980]. This uncertainty about the reliability of Australian data severely limits the utility of comparing Gondwanaland results with those from

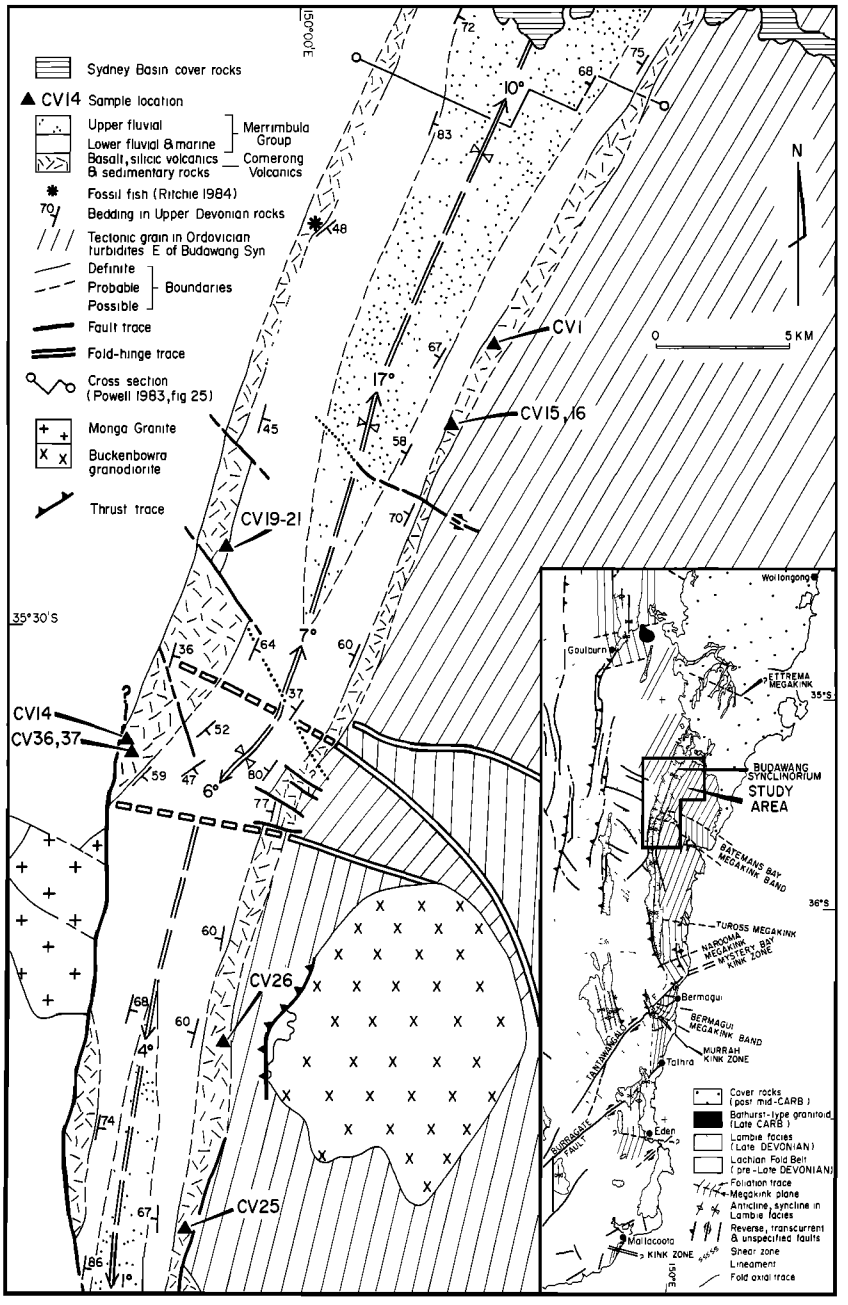


Fig. 1. Location and geological maps of the sampling area in the Budawang Synclinorium. Penetrative cleavage in the Ordovician turbidites predates accumulation of the Comerong Volcanics, but was kinked together with the Budawang Synclinorium in the mid-Carboniferous. The thrust faults marked are high-angle reverse faults of a few kilometers maximum dip-slip displacement.

Laurussia (North America and Europe west of the Urals) [Kent et al., 1984]. A major problem with the interpretation of many of the pole positions concerned is the paucity of field tests, such as the fold, contact or conglomerate tests [Graham, 1949]. This, coupled with the broad age constraints on many rock units naturally leads to equivocal conclusions. Although the contact and conglomerate

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

tests are commonly difficult to implement, the easy-to-use fold test has been much under-utilized in the past. The study of the Comerong Volcanics was undertaken because their age is tightly constrained and their structure is suitable for the application of a fold test. Definition of a prefolding magnetization direction in the volcanics also allows testing of the megakink hypothesis [Powell, 1984a; Powell et al., 1985].

GEOLOGICAL SETTING

The Comerong Volcanics is a compositionally bimodal suite of basaltic and rhyolitic volcanics, with intercalated sediments, which formed part of a late Middle or early Late Devonian continental rift in southeastern Australia [Fergusson et al., 1979; Powell, 1983a, 1984b]. The volcanics are overlain conformably by quartzose clastics of the Late Devonian Merrimbula Group and both units are preserved in the tight Budawang Synclinorium (Figure 1) [McIlveen, 1975; Powell, 1983a]. In the study area, the thickness of the Comerong Volcanics is variable; generally only a few hundred metres on the eastern limb [Dadd, 1984] and up to 2 km on the western limb of the Budawang Synclinorium (Figure 1). Lateral facies and thickness changes are abrupt, suggesting syndepositional movements, possibly on SE-trending faults [Dadd, 1984]. Basalt occurs at several levels in the volcanics (see columns A to C [Dadd, 1984] for representative sections) and has been sampled from both high and low stratigraphic levels in the formation, the choice of site depending largely on available outcrop. Rhyolites were also sampled but failed to provide any meaningful results and are not considered to be attractive palaeomagnetic targets.

The age of the Comerong Volcanics is well constrained to late Middle or early Late Devonian, as outlined below. The volcanics are younger than major transcurrent faults which cut the Early to Middle Devonian Bega Batholith (390 to 380 Ma discussed in Powell [1983a, b], and older than the late Frasnian Ettrema Limestone [Pickett, 1972], which is part of the marine lower part of the Merrimbula Group. A Rb/Sr age of 371 ± 6 Ma and a K/Ar age of 376 ± 6 Ma have been obtained from the alkali Monga Granite (Figure 1) considered to be comagmatic with the rhyolitic volcanics [Wyborn and Owen, 1982]. Fossil fish found in the southern correlative Boyd Volcanic Complex have been ascribed a late Givetian to early Frasnian age (G.C. Young in Fergusson et al. [1979]), and such an age is consistent with the nature of Phyllolepid fish remains uncovered from the Comerong Volcanics in the northwestern part of the study area [Ritchie, 1984]. There is thus good geological evidence that the Comerong Volcanics were erupted in a fairly brief interval around the Givetian-Frasnian boundary about 375-370 Ma ago.

STRUCTURAL CORRECTIONS

The Budawang Synclinorium is part of the regional meridional Early Carboniferous fold system in the eastern Lachlan Fold Belt [Powell et al., 1977]. The synclinorial structure extends more than 100 km in a N-S direction to the edge of the overlying Sydney Basin, and thence at least 50 km further NNE in the subsurface. The part of the Budawang Synclinorium from which the palaeomagnetic samples were collected is cut across by the ESE-trending Batemans Bay megakink band, a mid-Carboniferous structure in which part of the Budawang Synclinorium has been rotated by 19° clockwise with respect to the NNE-trending segment lying to the north (Figure 1) [Powell et al., 1985].

The structural map (Figure 1) shows a culmination where the Batemans Bay megakink band crosses the Budawang Synclinorium. Thus, only the lower half of the >3-km-thick Merrimbula Group [Powell, 1983a, Figure 25] outcrops in the megakink band, and fold axes in the Budawang Synclinorium plunge north and south away from the kink zone. In restoring the palaeomagnetic data to their presumed prefolding attitude, we have used fold axes calculated locally from segments of the Budawang Synclinorium (Table 1). In the megakink band, we have used an additional rotation of 19° counter-clockwise about a vertical axis in order to bring the palaeomagnetic data into alignment with the structural domain to the north (compare Domains A and B of Powell et al. [1985, Figures 10 and 12 and Table 1]). The procedure we have used in restoring the palaeomagnetic vectors to their prefolding attitude is first to remove the fold plunge, and, secondly, the unplunged bedding tilt.

All the sample sites are in basalt of

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Structural Corrections Used to Restore Palaeomagnetic Vectors

Site	F_1	α_{95} a	b	S_o	Estimated Error	Location
CV01	17/015	5.0	1.2	70/290	±5°	Belowra Ck., GR 08672884 Brooman
CV14	*First 19° counter-clockwise, about a vertical axis, then 6/201†	4.5	2.5	59/134	±10°	Mongarlowe R., GR 65286308 Monga
CV15	17/015	5.0	1.2	60/290	±5°	Bimberamala Ck., GR 06842552 Brooman
CV16	17/015	5.0	1.2	60/290	±5°	Bimberamala Ck., GR 06842552 Brooman
CV19	7/022	2.8	2.2	63/101	±5°	Currowan Ck., GR 69397065 Braidwood
CV20	7/022	2.8	2.2	63/101	±5°	Currowan Ck., GR 69377075 Braidwood
CV21	7/022	2.8	2.2	63/101	±5°	Currowan Ck., GR 69377075 Braidwood
CV25	1/184	12.6	2.8	68/273	±4°	Quart-Pot Ck., GR 66914480 Araluen
CV26	4/194	2.3	1.4	64/283	±6°	Branch of Buckenbowra R., GR 68735185 Araluen
CV36	*First 19°	4.5	2.5	59/134	±10°	Mongarlowe R., GR 65576281 Monga
CV37	counter clock-wise about a vertical axis, then 6/201†	4.5	2.5	59/134	±10°	Mongarlowe R., GR 65576281 Monga

Grid References (GR) are to 1:25,000 topographic maps, Central Mapping Authority, Bathurst, New South Wales.
*This rotation is estimated from the data herein and that given in Powell et al. [1985, Figures 10 and 12, and Table 1].
a and b are the major and minor semiaxes of the elliptical cone of 95% confidence calculated by the method described by N. I. Fisher [Appendix in Powell et al., 1985]. In each case, the greatest uncertainty (major semiaxis) is in plunge angle.
†This fold axis has been rotated 19° counterclockwise from its present-day orientation of 6/220 [Powell et al., 1985, Domain B].

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the Comerong Volcanics. Bedding determinations within the Comerong Volcanics are sparse, and generally not determined more accurately than ±10°, though sufficient are available to confirm the general paraconformable relationship to the Merrimbula Group inferred on stratigraphic grounds [Powell, 1983a]. Accordingly, we have used the orientation of bedding in the immediately overlying parts of the Merrimbula Group to give the best estimate of regional bedding at each site (Table 1). The accuracy quoted ("estimated errors") takes into account instrumentation error and variability between nearby outcrops. Most of the possible error is in the dip, as the strike is generally well constrained to ±2°.

TECHNIQUES

Routine procedures [Collinson et al., 1967] were followed in the collection and preparation of oriented rock specimens, nominally 25 mm diameter x 22 mm high. Both drill core and block samples were used with two or three specimens being cut from each. Magnetic measurements were performed with a μ-metal shielded three-axis CTF cryogenic magnetometer interfaced to a computer/disc storage system for on-line data handling. The magnetometer is housed in a 4-m cube three-axis Helmholtz coil system which serves the dual purpose of assisting the μ-metal shields to zero the ambient magnetic field in the sensor region during cool-down time, prior to the superconducting shield becoming operative, and of maintaining a low magnetic field (<500 nT) environment at the specimen insertion level after cool-down with the appropriate manual adjustment of the field controls. The system has no cryo-cooler to extend the helium refill period, but this is offset by the considerable advantage in keeping a low field at the magnetometer head. Helium gas recycling is achieved by compressing and reliquifying boil-off gas at the nearby CSIRO National Measurement Laboratories. Demagnetisation was effected by a custom-built nonmagnetic automatic furnace located in a low magnetic field controlled by feedback within ±5 nT (±5 γ) [Coward et al., 1985] or a Schonstedt GSD1 alternating field (A.F.) demagnetiser housed inside μ-metal shields. Specimens are transferred from demagnetising stations to the magnetometer inside μ-metal shields ensuring that they are exposed to minimal fields once treatment has begun.

All data are stored on disc for fast retrieval and transfer to a Tektronix 4051 Graphics System which displays demagnetization data in the form of orthogonal plots [Zijderveld, 1967] and stereonet plots for immediate evaluation. Following this, a number of analytical procedures are available depending upon the degree of complexity displayed. Initial reduction of the data involves processing with principal component analysis (PCA) [Kirschvink, 1980] using a 10° maximum angular deviation (MAD) cutoff. From the orthogonal plots and the PCA the multicomponent nature of the magnetizations present in these volcanics was apparent. Both magnetite and haematite are major remanence carriers. The selection criteria applied to the PCA directions was designed to eliminate spurious components as far as possible due to viscous remanent magnetization (VRM) or isothermal remanent magnetization (IRM). High natural remanent magnetization (NRM) intensities indicate that lightning has played a large role in magnetizing some of the samples collected. Therefore, only components stable above at least 500°C have been considered, and, when AF demagnetization was used, only the components of highest coercivity have been considered. To ensure objectivity and to increase the quality of the result, the data have been passed through a final filter, namely, only sites for which at least three samples have yielded similar directions after applying the above selection criteria have been used for the statistical computations.

RESULTS

Cleaning Characteristics and Magnetic Properties

NRM intensities varied greatly from only several mAm^{-1} (10^{-6} emu cm^3) to several thousand mAm^{-1}. The high intensities are commonly attributable to the effects of lightning being associated with single magnetic components. Some NRM intensities were up to 10% of their saturation magnetic intensities. These samples are not considered further.

Of the remaining samples, those from the eastern limb showed somewhat different

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

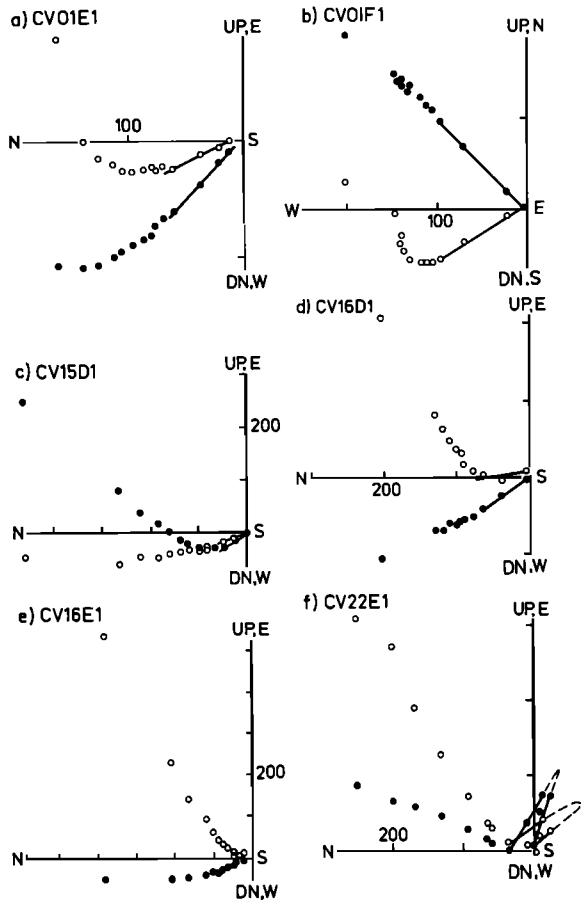


Fig. 2. Orthogonal projections of magnetisation vectors representing the thermal demagnetization of samples from the eastern limb. Closed (open) circles refer to the horizontal (vertical) plane while all units are in mAm^{-1} (x 10^{-6} emu cm^{-3}). Following the NRM step, the temperature steps (°C) used were 200, 300, 340, 390, 430, 480, 500, 520, 540, 550, 560, 570 and 580.

behaviour than those from the western limb. Figures 2a-2f and 3a-3f display orthogonal plots [Zijderveld, 1967] of the magnetization vectors of representative samples from the eastern and western limbs, respectively. All vectors are plotted with respect to present horizontal. The eastern limb samples showed simpler demagnetisation characteristics than those from the west. This was particularly so for samples from sites 01, 15 and 16 (Figures 2a-2e). These samples contain two main components and a third minor component of magnetisation. The low unblocking temperature component is usually close to the direction of the present geomagnetic field at the site, and, being destroyed by 300°-400°C or so, we interpret this component to be a viscous remanent magnetisation (VRM). The second component directed to the north-west and inclined shallowly down typically occupies the blocking temperature spectrum from about 300°-400°C, up to about 580°C, the Curie temperature of magnetite. That this magnetisation, and probably the less stable VRM, are carried by magnetite, as their stability ranges suggest, is supported by the observed susceptibility

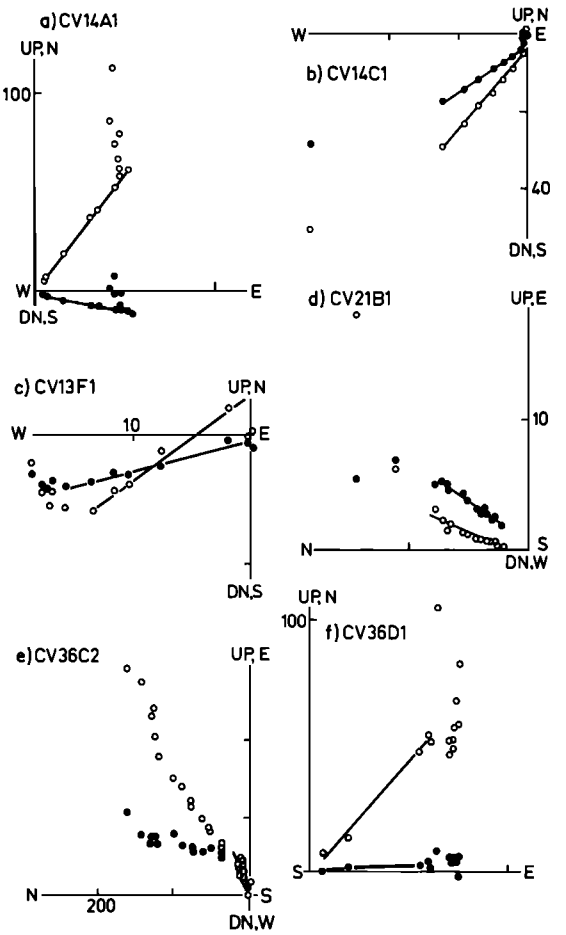


Fig. 3. Orthogonal projections of magnetisation vectors representing the thermal demagnetization of samples from the western limb. Symbols and units are the same as for Figure 2. Temperature steps (°C) are 200, 350, 400, 450, 500, 550, 570, 580, 590, 600, 630, 640 and 660. Some samples were taken higher to 690°C.

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

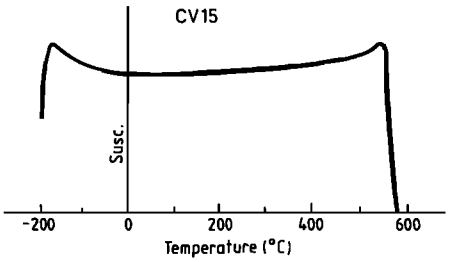


Fig. 4. Susceptibility versus temperature variation for a sample of Comerong Volcanics from site CV15. The low temperature isotropic point appears slightly lower than it really is because of the temperature gradient present at such temperatures.

versus temperature curve (Figure 4). This curve from samples of site 15 displays the quintessential characteristics of pure magnetite often associated with high-fidelity magnetic signatures [Schmidt and Embleton, 1981, 1985]. The Hopkinson peak just below 580°C indicates a predominance of optimum grain sizes with high unblocking temperatures and thus long relaxation times, while the low temperature isotropic point clearly suggests the presence of multidomain magnetite.

While the magnetite components dominate the magnetisation of the samples from the eastern limb, a third component, with unblocking temperatures in excess of the Curie temperature of magnetite, is often apparent, since the demagnetization trends of the orthogonal projections are offset from the origin. This is particularly evident in Figures 2a, 2b and 2d. The direction of this third component was not well defined nor consistent from sample to sample. At some sites where this component was more highly developed, there was an indication that the component was opposed to the high-temperature magnetite component (Figure 2f). Moreover, the coincidence of the maximum observed unblocking temperature with the Curie temperature of haematite at about 680°C, suggests that the third minor component is associated with haematite. Thin-section examination reveals that haematite occurs as staining along grain boundaries and cracks, suggesting it is secondary and may have therefore formed over a long period of time, or at least sometime after initial cooling (K. Dadd, personal communication, 1985). This may account for the polarity differences of the magnetite and haematite seen in samples from site 22 (Figure 2f), and the inconsistent directions of the haematite seen in other sites.

Samples from the western limb generally were less intensely magnetized than those from the eastern limb and commonly displayed behavior like the haematite magnetizations from the eastern limb. This, plus the much more altered appearance in hand specimen of the western limb samples, suggests the greater development of haematite in this limb. Maximum unblocking temperatures observed are well in excess of the magnetite Curie point and more consistent with that of haematite. Within-site reversals of magnetization directions are common (Figures 3a and 3b), indicating that not all samples within the one flow unit have been magnetized at the same time. There is also evidence of within-sample reversals, although they are not fully resolved by cleaning (Figure 3c; - the relatively low intensity of this sample is taken as evidence that this sample contains only haematite). Orthogonal projections of magnetization vectors for samples from the western limb are less smooth and commonly erratic compared with those from the eastern limb (see Figures 2 and 3). Although samples from the western limb are generally weakly magnetized, the erratic demagnetization trends are not associated with the sensitivity available for measurement, it is several orders of magnitude greater than that required to measure even the most weakly magnetized sample encountered here. It is far more likely that the haematite magnetizations are complex and have formed over a considerable period of time during which the ambient field direction would have changed, either due to secular variation and/or reversals. This is consistent with the above conclusions regarding the haematite in samples from the eastern limb. This strongly suggests that the haematite is secondary, or diagenetic but does not rule out the possibility that it formed soon after cooling, before the beds were folded, in which case it may be used legitimately to compare with magnetite results from the eastern limb. In fact, there are some sites from the western limb from which both magnetite and haematite components appear to have been identified and which have similar directions (CV19-21). These have been averaged for the

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Summary of Magnetisations and Pole Positions From the Comerong Volcanics

		In Situ*		After All Corrections†			Pole Position‡	
Site	N	Dec (°)	Inc (°)	Dec (°)	Inc (°)	k	Lat (°S)	Long (°E)
CV01	6	319.9	14.9	343.9	-44.4	61.7	73.3	269.9
CV14§	5	077.4	-49.3	347.4	-44.4	27.5	75.7	277.6
CV15	5	332.2	15.0	347.8	-29.0	71.9	67.2	298.3
CV16	7	329.6	7.9	351.3	-36.2	80.3	72.8	301.3
CV19	6	040.1	-11.3	010.4	-30.7	151	68.9	358.7
CV20	4	035.1	-12.2	007.1	-26.6	200	67.6	348.4
CV21	4	037.5	-9.2	011.3	-27.4	197	66.7	358.9
CV25	4	152.1	21.9	194.0	35.2	33.2	69.7	011.2
CV26	3	137.2	18.0	183.4	57.5	18.3	86.2	105.0
CV36§	4	078.8	-43.5	355.6	-45.4	67.0	80.6	305.2
CV37§	4	078.7	-34.5	008.2	-44.2	143	78.1	008.6
	11	10.9	-20.8	000.4 (345.4)	-38.7 (-38.7)‖		76.9 (71.3)	330.7 283.1)‖

N, number of unit vectors; R, resultant of N unit vectors; k, K, precision parameters for directions and poles, respectively; α_{95}, A_{95}, the half angle of the 95% cone of confidence for the mean direction and pole position, respectively [Fisher, 1953].
* R = 7.346; k = 2.7; α_{95} = 34.3°.
† R = 10.742; k = 38.8; α_{95} = 7.4°.
‡ R = 10.759; K = 41.5; A_{95} = 7.2°.
§ These sites are from within the megakink zone and have been appropriately corrected as described in the text and Table 1.
‖ This direction and pole position refer to a possible 15° rotation of the Bundawang Synclinorium as described in the text.

purpose of analysis. Thus the magnetic mineralogical differences between the western and eastern limbs are more of degree than of kind.

The above features were also apparent from low-temperature cycling (in zero field) of selected samples that had been saturated in magnetic fields of around 1.2 T (12,000 Oe). These samples displayed a continuum of responses, from those of site 15 on the eastern limb to those of site 37 on the western limb. Saturation magnetizations were 0.2 Am^{-1} (emu cm^{-3}) and 0.01 Am^{-1}, respectively. After cycling to dry-ice temperatures (-80°C), neither of these intensities changed more than a few percent but after cycling to liquid nitrogen temperatures (-196°C) the site 15 value was reduced to about 0.1 Am^{-1}, reflecting the predominance of multidomain magnetite and the effect of the isotropic point at -150°C in samples from the eastern limb. The liquid nitrogen cycle affected the magnetisation of the site 37 sample minimally, reflecting the predominance of haematite. The reason for the dry-ice cycle having minimal effect on the haematite is thought to be related to the finely divided nature of the haematite and the high-stress anisotropy providing a resilience to the Morin transition (at -20°C) displayed by coarser haematite. Nevertheless, these crude low temperature experiments on IRM's broadly parallel the demagnetisation results of the NRM's of these rocks.

Table 2 summarizes results of all sites that have satisfied the selection criteria set out in the previous section, and preempts the following section (to

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

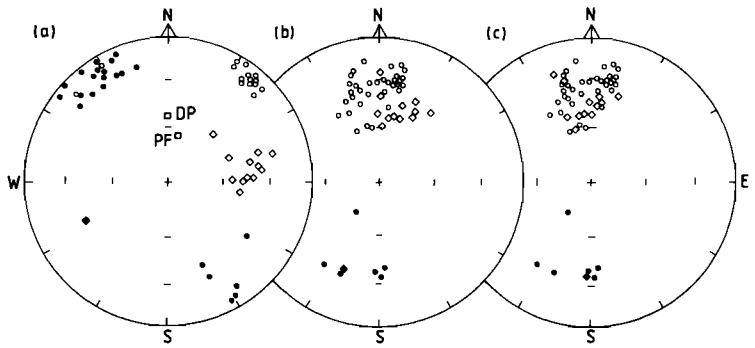


Fig. 5. Equal-area stereographic projections of cleaned sample magnetization directions: (a) in situ, (b) after correcting for plunge and folding, and (c) after correcting for proposed kinking [Powell et al., 1985]. Open symbols are up, closed symbols are down; diamonds are from within the megakink band, and circles from outside. DP and PF refer to the dipole and present magnetic fields at the sampling localities.

conserve space) by giving directions and pole positions after all structural corrections have been applied.

Structural Corrections of Magnetisation Directions

A comparison of directions before and after structural correction (Table 1) is given in Table 3 and displayed in Figure 5. The structural corrections described above are complicated by the extra rotation about a vertical axis of sites within the megakink band.

From the stereonet comparisons (Figure 5) there is clearly an improvement in grouping after correcting for the plunge and folding (Figures 5a and 5b), thereby obviating the need for any statistical test. However, a statistical appreciation of the improvement after unkinking is desirable because there is little

TABLE 3. Summary of Statistics Before and After (1) Plunge/Fold Correction and (2) Kink Correction

Sites	N	Dec	Inc	R	k	α_{95}
All before (1)	11	010.9	-20.8	7.345	2.7	34.3
All after (1)	11	005.2	-38.9	10.702	34.0	7.9
Outside kink	8	001.5	-36.3	7.738	33.0	9.8
In kink before (2)	3	016.1	-45.0	2.983	117	11.4
In kink after (2)	3	357.1	-45.0	2.983	117	11.4
All after (1) and (2)	11	000.4	-38.7	10.742	38.8	7.4

N, number of unit vectors; R, resultant of N unit vectors; k, precision parameter; and α_{95}, the half angle of the 95% cone of confidence for the mean direction [Fisher, 1953]. Statistics quoted are for rotating only sites within the megakink band to a common direction represented by the Budawang Synclinorium north of the megakink band. We have not attempted to carry out minor rotations between differing fold axis trends outside the megakink band, because this is probably beyond the precision of the palaeomagnetic data. Similarly, we have not attempted to rotate data from sites CV25 and 26 with respect to sites to the north of the megakink band because of the small amount of data and the lack of a known original direction to which all megakink domains can be referred. See text for discussion.

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

discernable difference between the distributions shown in Figures 5b and 5c. This was attempted using the test described by McFadden and Jones [1981] adapted to kinking rather than folding. In their notation and from our Table 3 we have the following:

$$N_a = 8 \quad R_a = 7.788 \quad k_a = 32.95$$
$$N_b = 3 \quad R_b = 2.983 \quad k_b = 117.28$$

where N is the number of mean site unit vectors, R is their resultant and k is the precision parameter [Fisher, 1953]. The critical value of the F-distribution (with $2(N_a-1) = 14$ and $2(N_b-1) = 4$ degrees of freedom) is 5.87 at the 95 percent confidence level. Since $k_b/k_a = 3.56$ there is no reason to suppose that the populations within and outside the megakink do not share a common precision parameter. Thus, we may proceed to test the hypothesis of a common true mean direction before and after unkinking (Table 3). For two groups and N = 11, the 95 percent critical value for rejecting a common true mean direction is 0.395 [McFadden and Jones, 1981, Table 1]. Thus, we must have

$$\frac{R_a + R_b - R^2/(R_a + R_b)}{2(N - R_a - R_b)} > 0.395 \ (0.292)$$

to reject the hypothesis with 95 (90) percent confidence. Using N, R_a and R_b given above, the critical value for R is 10.68 at the 95 percent level (i.e., R must be less than 10.68 to reject a common true mean direction). Before unkinking, R = 10.702 and increases to 10.742 after unkinking, which indicates that we are not able to reject the hypothesis of a common true mean direction for either (Figures 5b and 5c) at the 95 percent level. However, the resultant does increase after unkinking confirming that structural correction for the megakink improves the clustering of site directions. Moreover, the critical value for R is 10.704 at the 90 percent level indicating that the hypothesis may be rejected for the kinked group at this level, but not for the unkinked group indicating a positive kink test at the 90 percent level. The stereonet representation of the magnetic directions shows clearly the improvement in clustering by adjusting for the 19° megakink rotation, and we feel the reason this is not showing up more clearly in the statistical test is because the 19° rotation from the kinked to unkinked position is too small to change significantly the shape of the overall population of directions in which α_{95} is around 7°. The fact that there is such a marked improvement in clustering when comparing the unfolded directions with the present-day directions tells us that magnetisation predates the Early Carboniferous folding [Powell et al., 1977; Powell, 1983a]. Further, since megakinking structures are younger than the north-south trending folds [Powell 1983a; Powell et al., 1985], we are confident that our calculation of the pole positions given in Table 2, after corrections for folding, plunging and megakinking, provides the best estimate of the palaeomagnetic pole from our data. Because the magnetizations are prefolding, they have been acquired during accumulation of the Comerong Volcanics, about 370 Ma ago or shortly afterwards.

DISCUSSION

The coexistence of magnetite and haematite in most samples has contributed to the complexities encountered here but the simple pattern of results from samples whose behavior is dominated by magnetite (Figures 2 and 4) lends a good deal of confidence to the selection criteria adopted. These selection criteria have tended to be conservative (i.e., rejecting all sites within which less than at least three samples showed similar directions with comparable stabilities) and have been applied impartially. The ensuing directions undoubtedly satisfy a fold test (Figures 5a and 5b) and support the megakink hypothesis [Powell, 1984a; Powell et al., 1985]. The mean site direction is Dec = 0.4°, Inc = -38.7° and α_{95} = 7.4° with a mean site pole position of 76.9°S, 330.7°E and A_{95} = 7.2°. This pole is plotted in Figure 6a, where it is compared with other Australian pole positions thought to be of Silurian or Devonian age.

The poles available prior to 1980 were interpreted in a number of ways depending upon whether southeastern Australia, or the Lachlan Fold Belt (LFB) in particular, is considered as allochthonous. Figure 6a displays Australian poles for the Late Palaeozoic. The idea of southeastern Australia being allochthonous, or a displaced terrane, and being welded to the Australian craton during the Devonian,

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

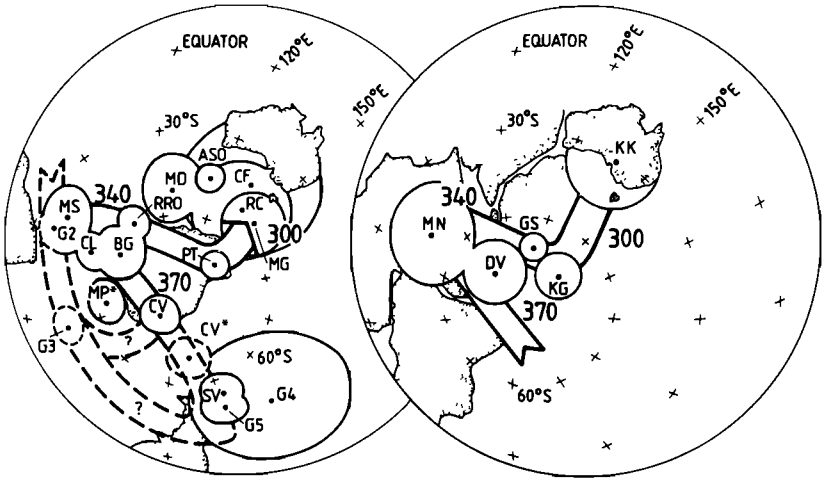


Fig. 6. Equal-area stereographic projections of Late Palaeozoic pole positions for (a) Australia, and (b) Africa rotated to Australia using rotation pole of Konig [1981] for Antarctica to Australia (1.5°N, 37.0°E) and a marginally looser rotation angle of 27.9°, as suggested by J. J. Veevers (personal communication, 1985). Norton and Sclater's [1979] pole was used for Africa to Antarctica (2.4°S, 327.3°E, -55.4). Pole mnemonics are those used by Embleton [1981], or as described in the text. CV* is the alternative Comerong Volcanics pole if the Budawang Synclinorium has been rotated 15° clockwise with respect to the Australian craton. See text for discussion. MP* is the structurally corrected pole discussed by Goleby [1980].

resulted from attempts to reconcile the apparent polar wander path (APWP) from southeastern Australia with the APWP from the geologically more mature central and western areas of Australia [McElhinny and Embleton, 1974; Embleton et al., 1974]. Alternatively, by invoking the "polarity option" for Early Palaeozoic poles, it has been shown to be possible to interpret the results as a single APWP [Schmidt and Morris, 1977], thus favoring the in situ development of the LFB. Critical to this interpretation is the age of the magnetization of the Mereenie Sandstone [Embleton, 1972], as discussed further below.

Another single-path interpretation by Morel and Irving [1978] accommodates the poles from southeastern Australia by expanding the mid-Palaeozoic section while retaining the original polarity assignations. Subsequently, Goleby [1980] reported new results from Early Palaeozoic rock formations from southeastern Australia and, after taking into account new age and structural information regarding some old results, synthesised another APWP. This last path also supports the in situ tectonic development of southeastern Australia, rejecting the allochthonous model. The last path is at odds with the APWP suggested by Schmidt and Morris [1977] in that the original polarities of the Early Palaeozoic poles are retained. In this aspect, Goleby's [1980] path is similar to Morel and Irving's [1978] APWP, although the timing of the paths is different. Goleby [1980] asserts that magnetic overprinting during the Devonian has reset some Silurian rocks and indicates the need for more work to test this. However, when the APWP is examined in detail several anomalies persist. Goleby's Early-Middle Devonian pole position, plotted as G4 in Figure 6a, has a large error associated with it, and since the precision of the mean direction decreases further (from 16 to 11) when the directions used to calculate this pole position are corrected for bedding tilt [Goleby, 1980, Table 1], there is the probability that these rocks have themselves been remagnetized. Thus, it is premature to reassign the Silurian Volcanics (SV) pole [Luck, 1973] and the Ainslie Volcanics (AV) [Luck, 1973] as replotted by Goleby [1980], to the Devonian. Yet if the Silurian pole positions of Goleby [1980] are valid (G2 and G3, Figure 6a), which is supported by

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the fold test he reports, then the SV pole appears to be quite divergent. In any case, if all the pole positions plotted in Figure 6a are real estimates of the pole position sometime in the Late Palaeozoic, be they pre- or postfolding, then a large amount of polar wandering relative to Australia (and maybe Gondwanaland) is indicated. This is at variance with results from similarly aged rocks of Laurussia [Van der Voo, 1981].

The value of the results from the Comerong Volcanics (Table 2) is that they satisfy a fold test, and the age of the volcanics is tightly constrained to the latest Middle or earliest Late Devonian. They are paraconformably overlain by the quartzose clastic Lambie facies, which can be correlated by palaeocurrent and provenance relationships over the entire LFB in the Late Devonian [Powell, 1984b]. There are no known disjuncts in the facies or palaeodrainage pattern, nor any justaposition of contrasting Late Devonian terranes, so that we are confident there has been no large displacement or rotation of the Comerong Volcanics with respect to the Australian craton. A small rotation of the Budawang Synclinorium north of the Batemans Bay megakink band by 15° clockwise with respect to the Australian craton may be sustained if the 030° trend of the tectonic grain in the Ordovician turbidites has been rotated from a premegakinking orientation of 015° (see Powell, 1984a, Figure 2). We do not have sufficient palaeomagnetic data to test this hypothesis, and accordingly have used a megakink correction only for the sites within the Batemans Bay megakink band (Figure 1; Tables 1 and 2) to determine our preferred Comerong Volcanics pole (76.9°S, 330.7°E, A_{95} = 7.2°). If the northern part of the Budawang Synclinorium is rotated by 15° clockwise from its original Early Carboniferous attitude, then the alternative pole for the Comerong Volcanics is 71.3°S, 283.1E, A_{95} = 7.2°. As shown in Figure 6a, this alternative pole is not greatly different from our preferred Comerong Volcanics pole, and lies on our postulated Australian APWP. Consequently, we think the Comerong Volcanics pole (CV, Figure 6a) is representative for Australia as a whole, and presumably for Gondwanaland.

Accepting that there is no evidence of large-scale relative displacement of crustal units within the LFB since the Middle Devonian, we do not find a consistent picture emerging when the Comerong Volcanics pole is compared with published results from Australian rock units of similar age. For instance, there is a large difference between poles CV and MD (Mulga Downs Group [Embleton, 1977] which is a correlative of the Late Devonian Lambie facies). This can be reconciled if the magnetization of the Mulga Downs Group is Carboniferous rather than Late Devonian, and was acquired during the Kanimblan folding which affected the entire LFB. We also note that, prior to tilt correction, the normal and reversed magnetizations from the Mulga Downs Group are reminiscent of Tertiary remagnetizations commonly associated with weathering in Australia [Schmidt and Embleton, 1976, Schmidt et al., 1976]. Without a field test, e.g., a fold test, it is not possible to distinguish when the magnetization was acquired. This problem is symptomatic of the difficulties faced when interpreting many of the LFB data, and we are led toward a cautious approach and consider that magnetizations may be secondary unless there is good evidence to the contrary.

In Figure 6a the Comerong Volcanics pole represents a key pole position for the Middle/Late Devonian (~370 Ma). For younger times the pole shifts northward from CV to BG (Early Devonian Bowning Group; Luck [1973]), now assumed to be secondary in the absence of evidence to the contrary, RRO (which was interpreted as secondary by Kirschvink [1978]) and to the MS pole and ASO (also secondary; Klootwijk [1980]). The assumed Siluro-Devonian age of the magnetization of the Mereenie Sandstone (MS) must now be called into question since the MS pole is close to the RRO pole. Both the rock units at Ross River (N.T.) and the Mereenie Sandstone, sampled at Ellery Creek (N.T.) 100 km west, have been strongly affected by the Alice Springs Orogeny during the Early Carboniferous (~340 Ma; Armstrong and Stewart [1975]). These poles are therefore assumed to date from about that time. A preliminary pole position has been calculated from new results from Late Devonian limestone reefs in northern Australia [Hurley and Van der Voo, 1985] and is plotted as CL in Figure 6a. In the absence of a fold test we can date this pole as being about 340 Ma, by association

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

with the RRO and MS poles. This is reasonable if the magnetization of the limestone is a chemical remanent magnetization (CRM) rather than a depositional remanent magnetization (DRM). We also note that the age of magnetization of the Housetop Granite, whose pole is not used here but is often quoted by others as a Devonian pole, is interpreted to be Tertiary [Briden, 1967].

The MD pole already discussed could be Early Carboniferous (by interpolation) or Tertiary if untilting is disallowed. The next group of poles represent some of the most reliable results available with convincing fold tests and very well documented demagnetization experiments indicating original magnetizations [Irving, 1966]. The poles from the Rocky Creek Formation (RC), The Currabubulla Formation (CF) and the Main Glacial Stage (MG) are key poles (Figure 6a) for the Late Carboniferous (300 ± 10 Ma, Herbert [1980]). Thus, Australia appears to have shifted from low latitudes (10°S-15°S) to high latitudes (70°S-80°S) in the time interval between 350 Ma and 300 Ma, and perhaps in an interval as short as 340 Ma to 310 Ma, as originally suggested by Irving [1966].

For times prior to the Late Devonian the pole path for Australia remains ambiguous because of uncertainty about the age of aquisition of magnetization at many sites. Various possibilities are dashed in Figure 6a. This problem is in urgent need of attention, and, although a detailed discussion is outside the scope of the present report, a reassessment of existing data in the light of some new Late Silurian/Early Devonian results is in preparation and should be reported shortly.

A comparison of the pole path suggested for times between 370 Ma and 300 Ma with African pole positions for these times is given in Figure 6b. Clearly these pole positions in general conform with the path, although their previously assigned ages are not entirely compatible. For instance, the Msissi Norite pole (MN) [Hailwood, 1974] was originally assigned to the latest Devonian (Famennian) on stratigraphic grounds, although a Carboniferous age could not be rejected. On our path the Msissi Norite would appear to be about 340 Ma or Early Carboniferous (Visean). Reconstruction details (given in the caption to Figure 6) use the rotations of Norton and Sclater [1979] for Africa to Antarctica and a slightly modified rotation of Konig [1981] for Antarctica to Australia. An alternative rotation of Powell et al. [1980] for Africa to Antarctica (or more generally, West Gondwanaland to East Gondwanaland) does not alter the comparison, principally because of the magnetization ages not being known with any certainty, and of the inherent looseness associated with errors on the pole positions, e.g., A_{95} = 16.5° for the Msissi Norite pole position.

The pole position from the Lower to Middle Devonian Gneiguira Supergroup (GS; Figure 6b) is more likely to relate to about 320 Ma (Namurian) than the Devonian. With the absence of a fold test, or other evidence to the contrary, this interpretation is clearly valid [Kent et al., 1984]. In fact, the existing evidence favors remagnetization because Upper Proterozoic to Cambrian (?) red beds from the same region in Africa (Mauritania) yield results similar to those from the Devonian red beds [Morris and Carmichael, 1978; Kent et al., 1984]. It seems likely that these rocks have been remagnetized simultaneously. In addition, the consistently reversed polarity observed in these rocks is suggestive of a Kiaman age. Note that the GS pole (Figure 6b) is very close to the Paterson transition [Irving, 1966], marked by the PT pole (Figure 6a). That red-bed units may be magnetized long after deposition has been often debated in the literature and the general consensus appears to be in favor of such late timing of aquisition [Roy and Park, 1974; Larson et al., 1982]. The interpretation of the poles KK and KG, from the K3 beds, is essentially that of McElhinny and Opdyke [1968], as is the interpretation of the Dwyka Varves (DV) pole. The pole position and mixed polarities of the varves are each consistent with a pre-Kiaman or Early Carboniferous age.

CONCLUDING REMARKS

We have presented palaeomagnetic results from the Comerong Volcanics which satisfy a fold test and support the megakink hypothesis for the Lachlan Fold Belt [Powell et al., 1985]. The pole position enables the calibration of the Australian pole path for Late Devonian

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

times (~370 Ma). Furthermore, we argue that the pole is representative for Gondwanaland as a whole. Many other Siluro-Devonian rock units are now thought to have been remagnetized, or to have acquired their magnetizations long after deposition. The Mereenie Sandstone pole position [Embleton, 1972], supported by the African Msissi Norite pole position [Hailwood, 1974] have been cornerstones of many interpretations of Siluro-Devonian palaeoreconstructions, but it now seems more likely that both these units have been magnetized in the Carboniferous. This interpretation is consistent with the geological history of these two units. We note that the pole path from the Late Devonian to the Late Carboniferous (~370 Ma to ~300 Ma) shows a considerably greater amount of motion than the Laurussian path, reflecting a more mobile Gondwanaland than previously thought.

Acknowledgments. This project was funded under the CSIRO/Australian Universities Collaboration Grant during 1984 and 1985. Mark Huddleston ably assisted in the collection and preparation of samples and is kindly thanked. We are grateful to John Veevers and Dave Clark for the careful reading and comments on our early manuscript. Chris Klootwijk and Rob Van der Voo are acknowledged for their suggestions to clarify many points.

REFERENCES

Armstrong, R. L., and A. J. Stewart, Rubidium-strontium dates and extraneous argon in the Arltunga Nappe Complex, Northern Territory, J. Geol. Soc. Aust., 22, 103-115, 1975.

Briden, J. C., Secondary magnetization of some Palaeozoic rocks from Tasmania, Pap. Proc. R. Soc. Tasmania., 101, 43-48, 1967.

Collinson, D. W., K. M. Creer, and S. K. Runcorn, Methods in Palaeomagnetism, 609 pp., Elsevier, New York, 1967.

Coward, D. A., G. S. Cripps, B. C. Maher, and B. H. Ridley, Zero magnetic field furnace controller, CSIRO Division of Mineral Physics 1983/84 Biennial Report, p.26, 1985.

Dadd, K., Volcanic stratigraphy and palaeogeography of the Eden-Comerong-Yalwal rift zone, Geol. Soc. Aust. Abstr., 12, 126-127, 1984.

Embleton, B. J. J., The palaeomagnetism of some Palaeozoic sediments from central Australia, J. Proc. R. Soc. N. S. W., 105 , 86-93, 1972.

Embleton, B. J. J., A late Devonian palaeomagnetic pole for the Mulga Downs Group, Western New South Wales, J. Proc. R. Soc. N. S. W., 110, 25-27, 1977.

Embleton, B. J. J., A review of the palaeomagnetism of Australia and Antarctica, in Palaeoreconstruction of the Continents, Geodyn. Ser. vol. 2, edited by M. W. McElhinny and D. A. Valencio, pp. 77-92, AGU, Washington, D.C., 1981.

Embleton, B. J. J., M. W. McElhinny, A. R. Crawford, and G. R. Luck, Palaeomagnetism and the tectonic evolution of the Tasman Orogenic Zone, J. Geol. Soc. Aust., 21, 187-194, 1974.

Fergusson, C. L., R. A. F. Cas, W. J. Collins, G. Y. Craig, K. A. W. Crook, C. McA. Powell, P. A. Scott, and G. C. Young, The Upper Devonian Boyd Volcanic Complex, Eden, New South Wales, J. Geol. Soc. Aust., 26, 87-105, 1979.

Fisher, R. A., Dispersion on a sphere, Proc. R. Soc. London Ser. A, 217, 295-305, 1953.

Goleby, B. R., Early Palaeozoic Palaeomagnetism in South East Australia, J. Geomagn. Geoelectr., Supp. III 32, 11-21, 1980.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, J. Geophys. Res., 54, 131-167, 1949.

Hailwood, E. A., Palaeomagnetism of the Msissi Norite (Morocco) and the Palaeozoic reconstruction of Gondwanaland, Earth Planet. Sci. Lett., 23, 376-386, 1974.

Herbert, C., Evidence for glaciation in the Sydney Basin and Tamworth Synclinorial Zone, A Guide to the Sydney Basin, edited by C. Herbert and R. Helby, Bull. Geol. Surv. N. S. W., 26, 274-293, 1980.

Hurley, N. F., and R. Van der Voo, Palaeomagnetism of Upper Devonian reefal limestones, Canning Basin, Western Australia, E-S Trans. AGU, 66, 254, 1985.

Irving, E., Palaeomagnetism of some Carboniferous rocks from New South Wales and its relation to geological events, J. Geophys. Res., 71, 6025-6051, 1966.

Kent, D. V., O. Dia, and J. M. A. Sougy, Palaeomagnetism of lower-middle Devonian and Upper Proterozoic-Cambrian (?) rocks from Mejeira (Mauritania, West Africa), in Plate Reconstruction from Palaeozoic

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Palaeomagnetism, Geodyn. Ser., vol. 12, pp. 99-115, AGU, Washington, D.C. 1984.

Kirschvink, J. L., The Precambrian-Cambrian boundary problem: Palaeomagnetic directions from the Amadeus Basin, central Australia, Earth Planet. Sci. Lett., 40, 91-100, 1978.

Kirschvink, J. L., The least squares line and plane and the analysis of palaeomagnetic data, Geophys. J. R. Astron. Soc., 62, 699-718, 1980.

Klootwijk, C. T., Early Palaeozoic palaeomagnetism in Australia, III, Middle to early-Late Cambrian results from the Amadeus Basin, Northern Territory, Tectonophysics, 64, 249-332, 1980.

Konig, M., Geophysical data from the continental margin off Wilkes Land, Antarctica: Implications for pre-drift fit and separation of Australia-Antarctica, E-S Trans. AGU, 62, 384, 1981.

Larson, E. E., T. R. Walker, P. E. Patterson, R. P., Hoblitt, and J. G. Rosenbaum, Palaeomagnetism of the Moenkopi Formation, Colorado Plateau: Basis for long-term model of aquisition of chemical remanent magnetism in red beds, J. Geophys. Res., 87, 1081-1106, 1982.

Luck, G. R., Palaeomagnetic results from Palaeozoic rocks of south-east Australia, Geophys. J. R. Astron. Soc., 32, 35-52, 1973.

McElhinny, M. W. and B. J. J. Embleton, Australian palaeomagnetism and the Phanerozoic plate tectonics of eastern Gondwanaland, Tectonophysics, 22, 1-29, 1974.

McElhinny, M. W. and N. D. Opdyke, Palaeomagnetism of some glacial varves from central Africa, J. Geophys. Res., 73, 689-696, 1968.

McFadden, P. L. and D. L. Jones, The fold test in palaeomagnetism, Geophys. J. R. Astron. Soc., 67, 53-58, 1981.

McIlveen, G. R., The Eden-Comerong-Yalwal Rift Zone and the contained gold mineralisation, Rec. Geol. Surv. N. S. W., 16, 245-277, 1975.

Morel, P., and E. Irving, Tentative palaeocontinent maps for the Early Phanerozoic and Proterozoic, J. Geol., 86, 535-561, 1978.

Morris, W. A. and C. M. Carmichael, Palaeomagnetism of some late Precambrian and lower Palaeozoic sediments from L'Adrar de Mauritanie, West Africa, Can. J. Earth Sci., 15, 253-262, 1978.

Norton, I. O. and J. G. Sclater, A model for the evolution of the Indian Ocean and the breakup of Gondwanaland, J. Geophys. Res., 84, 6803-6830, 1979.

Pickett, J. W., Late Devonian (Frasnian) conodonts from Ettrema, New South Wales, J. Proc. R. Soc. N. S. W., 105, 31-37, 1972.

Powell, C. McA., Geology of the N.S.W. south coast and adjacent Victoria, with emphasis on the pre-Permian structural history, Geol. Soc. Aust. Specialist Group in Tectonics and Structural Geology Field Guide, 1, 118 pp, 1983a.

Powell, C. McA., Tectonic relationship between the Late Ordovician and Late Silurian palaeogeographies of southeastern Australia, J. Geol. Soc. Aust., 30, 353-373, 1983b.

Powell, C. McA., Terminal fold-belt deformation: Relationship of mid-Carboniferous megakinks in the Tasman fold belt to coeval thrusts in cratonic Australia, Geology, 12, 546-549, 1984a.

Powell, C. McA., Uluru regime: (h) Late Devonian and Early Carboniferous: Continental magmatic arc along the eastern edge of the Lachlan Fold Belt, in Phanerozoic Earth History of Australia, edited by J. J. Veevers, pp. 329-340, Oxford University Press, New York, 1984b.

Powell, C. McA., D. R. Edgecombe, N. M. Henry, and J. G. Jones, Timing of regional deformation of the Hill End Trough: A reassessment, J. Geol. Soc. Aust., 23, 407-421, 1977.

Powell, C. McA., B. D. Johnson, and J. J. Veevers, A revised fit of East and West Gondwanaland, Tectonophysics, 63, 13-29, 1980.

Powell, C. McA., J. P. Cole, and T. J. Cudahy, Megakinking in the Lachlan Fold Belt, J. Struct. Geol., 7, 281-300, 1985.

Ritchie, A., A new Placoderm, Placolepis gen. nov. (Phyllolepidae), from the Late Devonian of New South Wales, Australia, Proc. Linn. Soc. N. S. W., 107, 321-353, 1984.

Roy, J. L. and J. K. Park, The magnetization process of certain red beds: Vector analysis of chemical and thermal results, Can. J. Earth Sci., 11, 437-471, 1974.

Schmidt, P. W. and B. J. J. Embleton, Palaeomagnetic results from sediments of the Perth Basin, Western Australia, and their bearing on the timing of regional lateritisation, Palaeogr.

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Palaeoclimatol. Palaeoecol., 19, 257-273, 1976.
Schmidt, P. W. and B. J. J. Embleton, Magnetic overprinting in south-eastern Australia and the thermal history of its rifted margin, J. Geophys. Res., 86, 3998-4008, 1981.
Schmidt, P. W. and B. J. J. Embleton, Prefolding and overprint magnetic signatures in Precambrian (~ 2.9-2.7 Ga) igneous rocks from the Pilbara Craton and Hamersley Basin, NW Australia, J. Geophys. Res., 90, 2967-2984, 1985.
Schmidt, P. W., and W. A. Morris, An alternative view of the Gondwana Palaeozoic apparent polar wander path, Can. J. Earth Sci., 14, 2674-2678, 1977.
Schmidt, P. W., D. T. Currey, and C. D. Ollier, Sub-basaltic weathering, damsites, palaeomagnetism and the age of lateritization, J. Geol. Soc. Aust., 23, 367-370, 1976.
Van der Voo, R., Paleomagnetism of North America: A brief review, in Paleoreconstruction of the Continents, Geodyn. Ser., vol. 2, edited by M. W. McElhinny and D. A. Valencio, pp. 159-176, AGU, Washington, D.C., 1981.
Wyborn, D., and M. Owen, Araluen geological sheet 8826, scale 1:100,000, Bur. of Miner. Res., Geol. and Geophys., Canberra, 1982.
Zijderveld, J. D. A., A.C. demagnetisation of rocks, in Methods in Palaeomagnetism, edited by D. W. Collinson, K. M. Creer and S. K. Runcorn, pp. 254-286, Elsevier, New York, 1967.

T. J. Cudahy and C. McA. Powell, Australian Plate Research Group, School of Earth Sciences, Macquarie University, North Ryde, New South Wales 2113, Australia.

B. J. J. Embleton and P. W. Schmidt, CSIRO Division of Mineral Physics and Mineralogy, P.O. Box 136, North Ryde, New South Wales 2113, Australia.

(Received May 23, 1985;
revised August 5, 1985;
accepted August 5, 1985.)

19449194, 1986, 1, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/TC005i001p00135 by Curtin University Library, Wiley Online Library on [25/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License